

Project Review



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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

SCHOOL OF ENGINEERING AND TECHNOLOGY

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Project Details

- **Project Title:** CMOS-Based 2:1 Multiplexer with XOR Gate for Fault-Detection
- **Team Members:**
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- **Guide:** Dr. Jitendra Kumar

Problem Statement & Objectives

Problem Statement:

The project aims to develop a **CMOS-based 2:1 multiplexer integrated with an XOR gate** that enhances real-time fault detection capabilities in modern computing systems. Traditional multiplexers lack built-in fault detection, which risks undetected errors particularly in critical and high-reliability systems like AI processors and quantum computing. The use of an XOR gate for comparing expected and actual outputs enables immediate detection of faults in multiplexer selections, improving reliability, signal integrity, and fault tolerance while maintaining low power and high speed with 90nm CMOS technology.

Objectives:

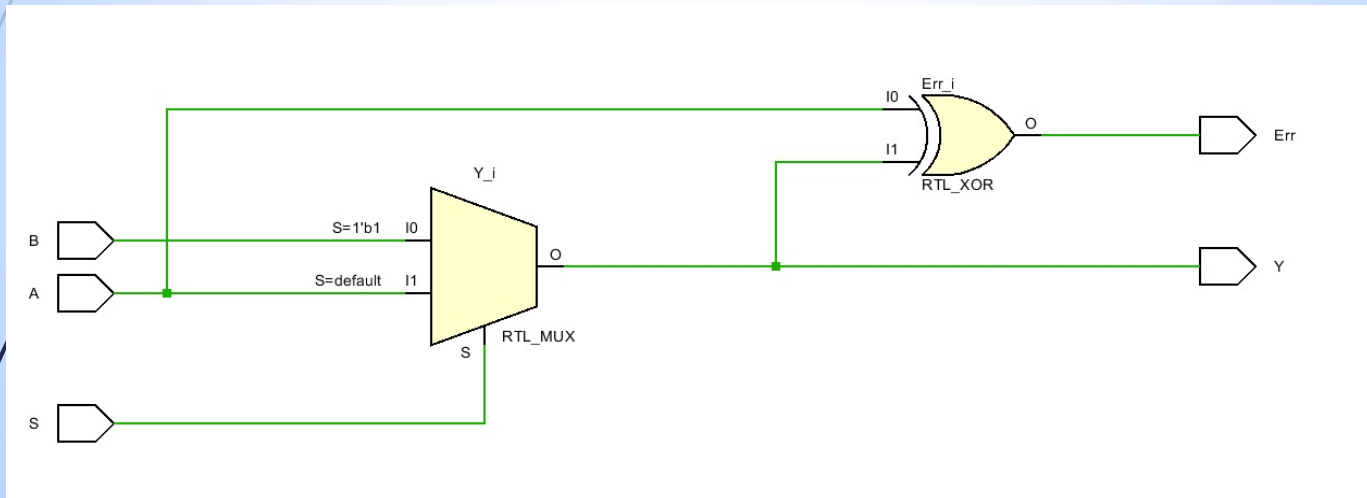
1. Design and implement a CMOS 2:1 multiplexer using transmission gates composed of complementary PMOS and NMOS transistors to ensure full voltage swing and low power consumption.
2. Integrate an XOR gate to serve both as a signal selector and real-time fault detector by comparing expected and actual outputs for error indication.
3. Achieve a design optimized for scalability and integration in fault-tolerant systems, particularly targeting AI hardware, quantum computing, FPGA systems, and low-power VLSI designs.

Literature Review

- Summarize key findings from related works:
- • Various 2:1 multiplexer design styles have been compared, focusing on power, delay, and area trade-offs, highlighting transmission gate CMOS designs as favorable for low power and robustness (ICATCT 2020).
- • CMOS transmission gates, employing both PMOS and NMOS simultaneously, ensure full voltage swing and reduce signal degradation compared to pass transistor-only designs, improving output signal integrity.

Proposed Methodology

- **Approach and Techniques:** Truth table, 90tech gpdk
- • **Software:** Schematic in EDA (Cadence Virtuoso) transient analysis.
- • **Tools/Platforms:** Cadence Virtuoso



Expected Outcomes

Anticipated results or deliverables:

- • A CMOS 2:1 multiplexer with an integrated XOR fault detection gate capable of reliable operation at 90nm technology node, showing 2 errors i.e. on 4th and 6th half cycle.
- • Enhanced fault detection capabilities that improve overall system reliability, especially in fault-sensitive environments like AI processors, FPGA systems, and quantum computing hardware.

Truth Table

This below is the truth table for the circuit.

- A and B are the inputs and s is select line of mux.
- Y(mux) is output of 2:1 mux and y(out) compares A and y(mux).
- At 4th and 6th cycle it detects error, so, y(out) is 1.

A	B	S	Y(mux)	Y(out)=y(mux) xor A
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	1	0
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

Schematic design

